Module 8: Building Visualizations in Power BI

📖 Overview

In the previous module, you learned the theory behind data visualization and dashboard design. In this module, you will apply those principles by creating visualizations in Power BI.

You will learn how to build, format, customize, and interact with Power BI visuals to transform raw data into meaningful insights.

This module focuses on the most commonly used visuals in real-world business reporting and dashboard development.

🎯 Learning Objectives

After completing this module, you will be able to:

- Create visualizations in Power BI
- Understand the Visualizations Pane
- Build charts and graphs
- Create KPI indicators
- Use tables and matrices
- Apply formatting and customization
- Configure tooltips and labels
- Create interactive reports
- Use slicers and filters effectively

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1️⃣ Introduction to Visualizations

What is a Visualization?

A visualization is a graphical representation of data that helps users:

- Analyze trends
- Compare values
- Identify patterns
- Make decisions

Why Visualizations Matter

Instead of reading thousands of rows:

Month	Sales
Jan	₹100,000
Feb	₹120,000
Mar	₹150,000

A chart immediately communicates the trend.

2️⃣ Visualizations Pane

The Visualizations Pane is the primary workspace for creating charts.

Main Sections

Visual Gallery

Contains available visuals.

Examples:

- Bar Chart
- Line Chart
- Pie Chart
- Table
- Matrix

Build Visual

Assign fields to visual components.

Format Visual

Customize appearance.

Analytics

Add:

- Trend Lines
- Forecasting
- Reference Lines

3️⃣ Building Your First Visual

Steps

Step 1

Load data into Power BI.

Step 2

Select a visual.

Example:

```
Clustered Column Chart
```

Step 3

Drag fields.

Example:

```
Category → Axis
Sales → Values
```

Step 4

Power BI generates the chart automatically.

4️⃣ Bar Charts

Purpose

Compare categories horizontally.

Example

Sales by Product.

Product	Sales
Laptop	₹50,000
Mobile	₹35,000

Types

Clustered Bar Chart

Categories displayed side by side.

Stacked Bar Chart

Displays contribution to total.

100% Stacked Bar Chart

Displays percentage contribution.

Best Use Cases

- Product Comparison
- Regional Comparison
- Ranking Analysis

5️⃣ Column Charts

Purpose

Compare categories vertically.

Types

Clustered Column Chart

Most common.

Stacked Column Chart

Shows category contribution.

100% Stacked Column Chart

Shows proportional contribution.

Best Use Cases

- Monthly Sales
- Quarterly Revenue
- Category Comparison

6️⃣ Line Charts

Purpose

Show trends over time.

Example

Monthly Revenue Trend.

Components

X-Axis

Time Dimension.

Example:

```
Month
```

Y-Axis

Measure.

Example:

```
Revenue
```

Best Use Cases

- Growth Analysis
- Trend Analysis
- Forecasting

7️⃣ Area Charts

Purpose

Highlight trends and magnitude.

Example

Monthly Sales Volume.

Advantages

Shows:

- Trend
- Contribution
- Volume

Best Use Cases

- Revenue Growth
- Website Traffic
- Customer Growth

8️⃣ Pie Charts

Purpose

Show part-to-whole relationships.

Example

Market Share.

Brand	Share
A	40%
B	30%
C	30%

Limitations

Avoid using:

- Too many categories
- Small differences

Recommended

Maximum:

```
5-6 categories
```

9️⃣ Donut Charts

Purpose

Alternative to Pie Charts.

Advantages

Can display:

- Total Value
- KPI

in center.

Example

Revenue Distribution by Category.

🔟 Tables

Purpose

Display detailed information.

Example

Customer	Revenue
Rahul	₹15,000
Priya	₹12,000

Features

- Sorting
- Filtering
- Drill-through

Best Use Cases

- Transaction Details
- Detailed Reports

1️⃣1️⃣ Matrix Visuals

Purpose

Cross-tab analysis.

Similar to Excel Pivot Tables.

Example

Region	Jan	Feb
North	100	120
South	150	180

Advantages

Supports:

- Row Hierarchies
- Column Hierarchies
- Drill Down

1️⃣2️⃣ Card Visuals

Purpose

Display a single KPI.

Example

```
Total Revenue

₹5.2M
```

Common KPIs

- Revenue
- Profit
- Customers
- Orders

Best Practices

Use large readable fonts.

1️⃣3️⃣ KPI Visuals

Purpose

Track performance against targets.

Components

Indicator

Current value.

Target

Expected value.

Trend

Historical performance.

Example

```
Revenue

Current: ₹5M
Target: ₹4.5M

↑ 11%
```

1️⃣4️⃣ Map Visuals

Purpose

Visualize geographical data.

Requirements

Location fields:

- Country
- State
- City
- Latitude
- Longitude

Types

Bubble Map

Displays values as bubbles.

Filled Map

Colors regions.

Example

Sales by State.

1️⃣5️⃣ Slicers

What is a Slicer?

A visual filter allowing users to interact with reports.

Example

Filter by:

- Year
- Region
- Product

Types

Dropdown

Compact format.

List

Displays all values.

Date Slicer

Filters time periods.

Benefits

Improves user experience.

1️⃣6️⃣ Filters

Power BI supports multiple filter levels.

Visual-Level Filters

Affect one visual.

Page-Level Filters

Affect all visuals on a page.

Report-Level Filters

Affect entire report.

Drillthrough Filters

Navigate to detailed pages.

1️⃣7️⃣ Tooltips

What are Tooltips?

Information displayed when hovering over visuals.

Example

Hover over a sales bar:

```
Sales: ₹50,000
Profit: ₹10,000
```

Benefits

Provides additional details.

1️⃣8️⃣ Conditional Formatting

Purpose

Highlight important values automatically.

Example

Profit

Positive:

```
Green
```

Negative:

```
Red
```

Applications

- Tables
- Matrix
- KPIs

Types

Background Color

Font Color

Data Bars

Icons

1️⃣9️⃣ Formatting Visuals

Formatting improves readability.

Common Formatting Options

Titles

Add meaningful titles.

Example:

```
Monthly Revenue Trend
```

Data Labels

Display values.

Legends

Identify categories.

Colors

Apply consistent themes.

Borders

Use sparingly.

Background

Maintain clean design.

2️⃣0️⃣ Report Interactivity

Interactive reports improve analysis.

Cross Filtering

Selecting one visual filters others.

Cross Highlighting

Highlights related data.

Drill Down

Move from summary to detail.

Example:

```
Year
 ↓
Quarter
 ↓
Month
```

Drill Through

Navigate to detailed report pages.

2️⃣1️⃣ Visualization Best Practices

Use Appropriate Charts

Choose based on business question.

Avoid Visual Clutter

Limit unnecessary elements.

Use Consistent Colors

Maintain uniformity.

Prioritize KPIs

Place at top.

Keep Layout Clean

Use white space effectively.

Limit Pie Charts

Use only when necessary.

📌 Key Terminologies

Term	Definition
Visual	Graphical representation of data
Axis	Horizontal or vertical chart dimension
Measure	Numeric value
Dimension	Category field
KPI	Key Performance Indicator
Slicer	Interactive filter
Tooltip	Hover information
Drill Down	Navigate to lower-level data
Cross Filter	Visual interaction
Conditional Formatting	Dynamic visual styling

📝 Practice Exercises

Exercise 1

Create:

- Bar Chart
- Column Chart
- Line Chart

Using sales data.

Exercise 2

Create a Card Visual displaying:

```
Total Revenue
```

Exercise 3

Build a Matrix showing:

```
Region × Month
```

Exercise 4

Create a Date Slicer.

Exercise 5

Apply Conditional Formatting to a Profit column.

🎯 Mini Project

Sales Performance Dashboard

Requirements

Create:

KPI Section

- Total Revenue
- Total Profit
- Total Orders

Trend Analysis

- Monthly Sales Trend

Category Analysis

- Revenue by Product Category

Geographic Analysis

- Sales by Region

Filters

- Year
- Region
- Product

Deliverables

1. Interactive Dashboard
2. Formatted Visuals

3. User-Friendly Layout
4. Meaningful Titles

📚 Module Summary

In this module, you learned:

✅ Visualizations Pane

✅ Bar Charts

✅ Column Charts

✅ Line Charts

✅ Area Charts

✅ Pie & Donut Charts

✅ Tables

✅ Matrix Visuals

✅ Card Visuals

✅ KPI Visuals

✅ Maps

✅ Slicers

✅ Filters

✅ Tooltips

✅ Conditional Formatting

✅ Report Interactivity

These visualization skills form the foundation of professional Power BI reporting. The next module introduces DAX, the calculation engine that powers advanced analytics and business metrics in Power BI.

⏭️ Next Module

Module 9: Introduction to DAX (Data Analysis Expressions)

Topics Covered:

- What is DAX?
- DAX Syntax
- Calculated Columns

- Measures
- Row Context
- Filter Context
- Basic DAX Functions
- Business Calculations
- Best Practices